Wages and Employment in a Single Labour Market

EC306- Wages and Employment

Justin Smith

Wilfrid Laurier University

Winter 2026

Goals of This Section

Goals of This Section

Today we will:

- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in the goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

Introduction

Introduction

- We have analyzed labour demand and supply separately in detail
- Equilibrium wage and employment is determined by the interaction of the two
- Market power plays an important role
  - In both goods markets and labour markets
- With a full understanding on how wages and employment are determined, we can analyze policies
- The minimum wage is one of the most prominent labour market policies
  - In theory, it leads to unemployment
  - The evidence is more mixed

Competitive Firms

Introduction

Assumptions

We start with firms that are competitive in both goods and labour markets

- Firms take the market wage as given
- They take the market price as given
- Competition means they are too small to influence either

- We analyze the equilibrium wage and employment for a homogeneous type of labour
  - All workers are the same

Market Demand and Supply

- The market demand and supply are the sum of the individual firms' demand and supply at each wage
- Here we determine the demand and supply curve for a single labour market
  - Could be a specific occupation in a specific region
  - Example: electricians in Ontario, professors in Canada, etc.
  - How the market is defined depends on context
  - But it is for a single type of labour

Market Demand and Supply

- Example is two firms with different elasticities of demand
  - Each firm demand curve is their value of marginal product
  - Firms may face different product prices if they sell different things (but hire the same labour)
- Each firm faces the same wage in the labour market
  - They perceive the supply curve as horizontal in the long run
  - They take the wage as given and can get as much labour as they want at that wage
  - Market supply is upward sloping
- Market demand is horizontal sum of individual demands
  - Add up the quantities demanded at each wage for each firm
  - Repeat for all wages

Market Demand and Supply

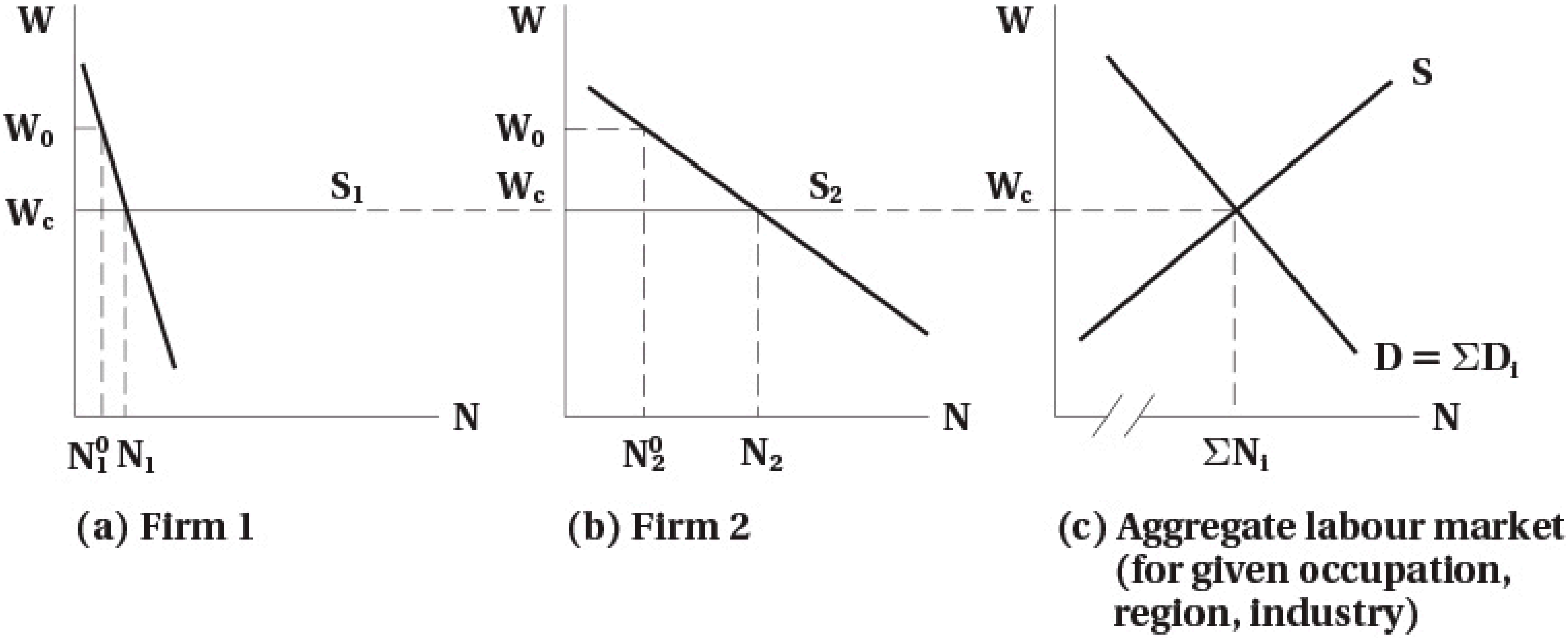


FIGURE 7.1 Competitive Product and Labour Markets

Firm Demand and Supply in the Short Run

- Above we analyzed the longer run
  - Firms can get as much labour as they want at the going wage
- In the short run, firms may face an upward sloping supply curve
  - They have to pay higher wages to attract more workers
  - This could be because of contracts, or other frictions
- If demand rises, they would pay higher wages to attract workers
- But because wages are high, over the long term more workers enter the market
  - Short run supply shifts out
  - Wage falls back down to long run level

Firm Demand and Supply in the Short Run

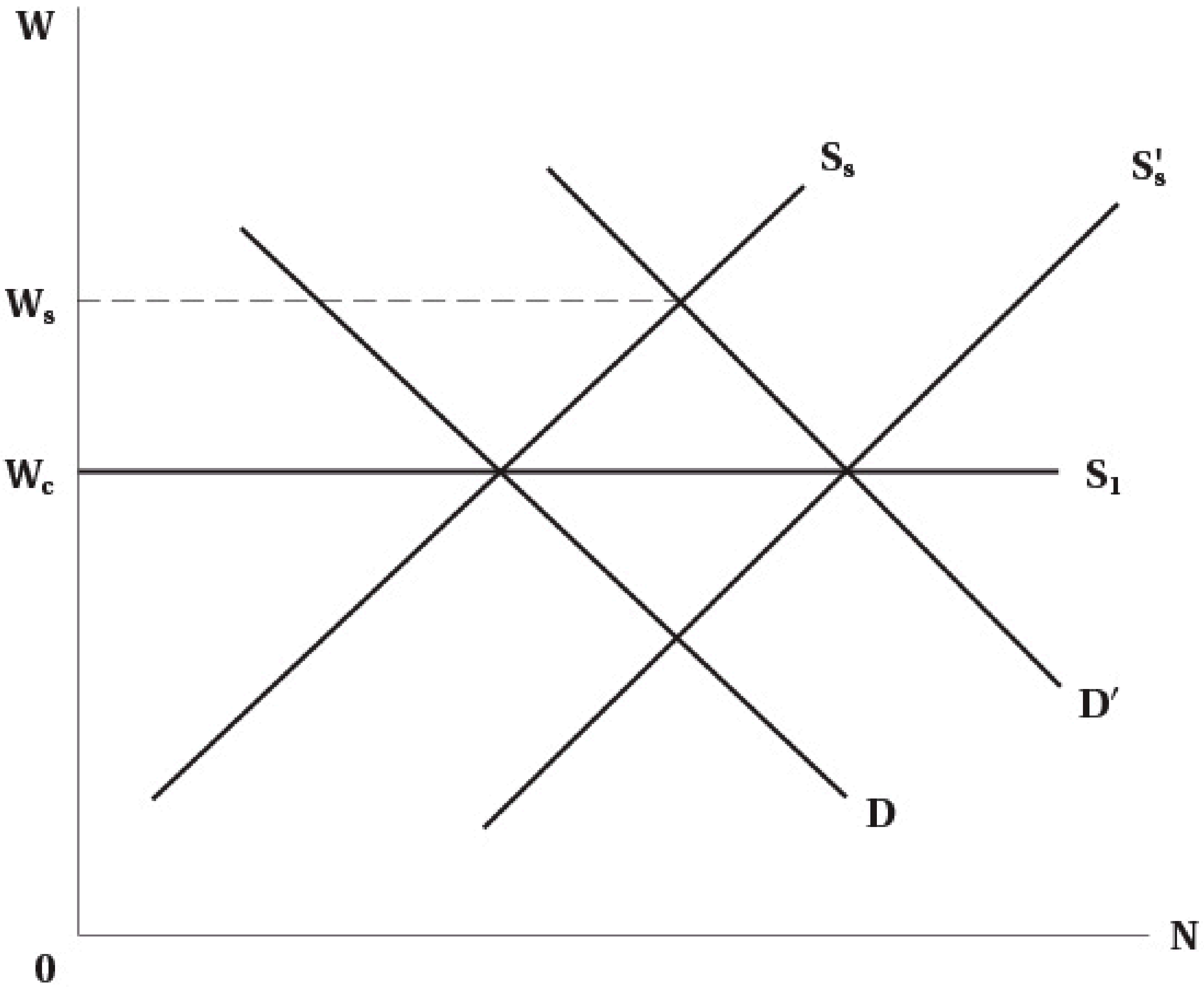


FIGURE 7.2 Short-Run and Long-Run Labour Supply Schedules to the Firm

Imperfect Competition in the Product Market

Monopoly

> **Monopoly**
>
> Monopoly: a single seller with market power — the ability to set prices above marginal cost

- For a monopolist, the labour demand curve is where wage equals marginal revenue product:

$$w = MR \times MPP_N$$

- For a monopolist, marginal revenue is less than price
  - Because to sell more, they have to lower price on all units sold
  - So marginal revenue product is less than value of marginal product (VMP)

Monopoly

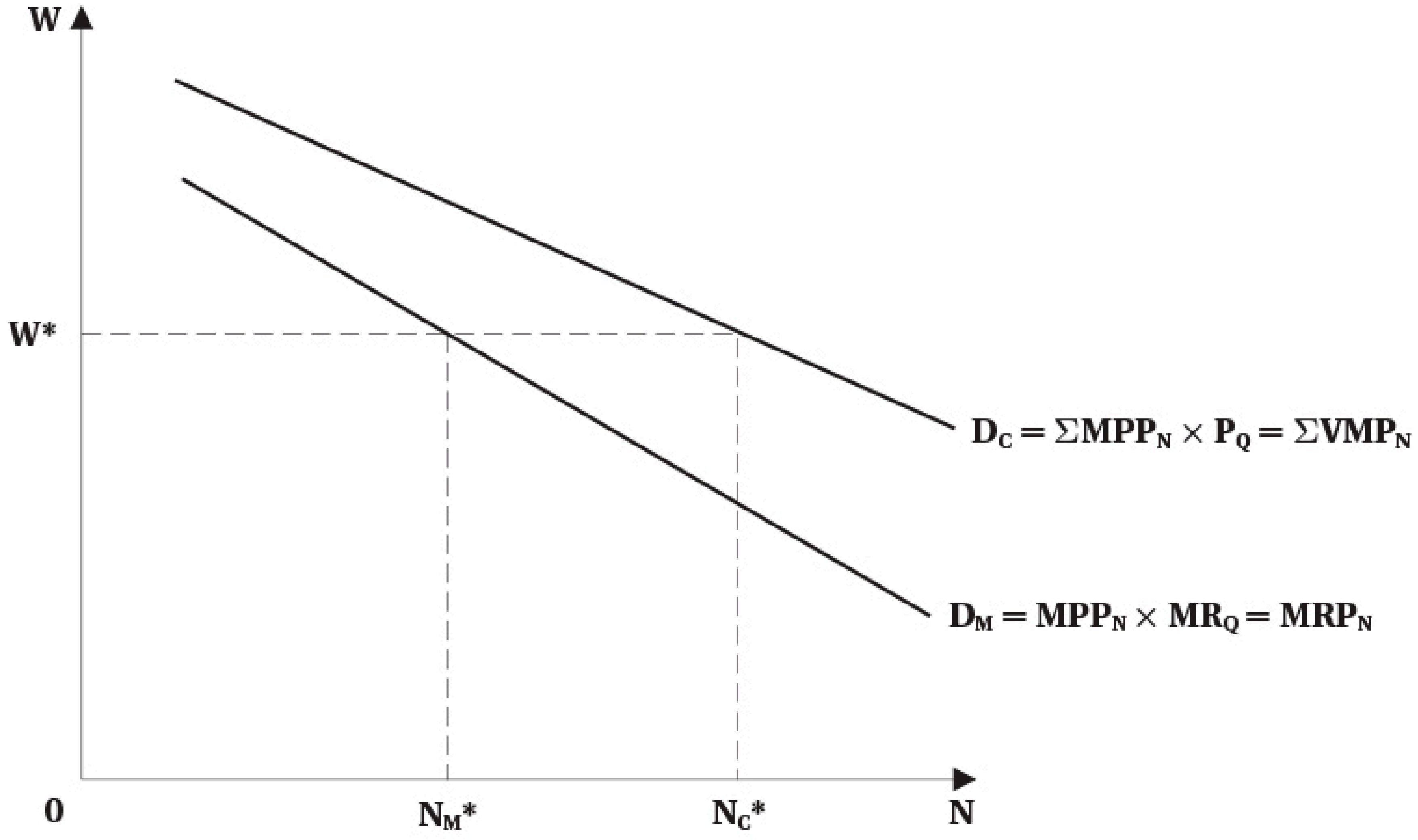


FIGURE 7.3 Monopolist versus Competitive Demand for Labour

Monopoly

> **Warning**
>
> Monopolists raise the price of their product above the competitive level, which decreases output — so they demand fewer workers than a competitive firm in the same market

- Note that power in the product market does not mean power in the labour market
  - The firm is still one of many hiring workers in a particular labour market

Monopolistic Competition and Oligopoly

- Many markets are neither perfectly competitive nor pure monopoly

Monopolistic competition:

- Many firms selling differentiated products
- Each firm has some market power because of product differentiation
- Can set prices above marginal cost

Oligopoly:

- A few firms dominate the market
- Each firm has significant market power
- But must consider reactions of competitors when setting prices

Monopolistic Competition and Oligopoly

Note

In both cases, demand for labour is based on marginal revenue product — which is less than VMP — so firms hire fewer workers than in competitive markets

Additional Details

- All of the previous analysis assumes that labour markets are perfectly competitive
  - Firms can hire all labour they want at the given wage
- That is not always the case in reality when firms have power in the product market
  - Employees may unionize
  - Firms may hire a large number of the market's workers
- In these cases, firms may not face a perfectly elastic supply curve for labour
- Later we will analyze imperfect competition in the labour market

Math of Labour Market Equilibrium

Introduction

- We know that market equilibrium is where demand and supply are equal
- You can do that graphically or mathematically
- In this section we cover both
- To make the math easier, we use linear demand and supply curves
  - This is just for simplicity — the same principles apply with non-linear curves

Demand and Supply Functions

Variables in the model:

- Demand: $N^D = f(w, Z)$
  - Z is a list of other factors affecting demand
- Supply: $N^S = g(w, X)$
  - X is a list of other factors affecting supply

Exogenous variables (Z and X) shift the curves:

- Examples: technology, prices of other inputs, preferences, demographics
- Determined outside the model

Endogenous variables (N^D, N^S, and w):

- Determined within the model

Demand and Supply Functions

Assume the demand and supply functions are linear:

- Demand: $N^D = a + bw$ (where a, b are constants)
- Supply: $N^S = c + fw$ (where c, f are constants)

To find equilibrium, set demand equal to supply:

$$a + bw^* = c + fw^*$$

Demand and Supply Functions

Equilibrium Solutions

Equilibrium wage:

$$w^* = \frac{c - a}{b - f}$$

Equilibrium employment:

$$N^* = \frac{bc - af}{b - f}$$

Use these equations to analyze how changes in exogenous variables affect equilibrium:

- Changing a shifts the demand curve (e.g., increase in product price)
- Changing c shifts the supply curve (e.g., increase in population)

Demand and Supply Functions

To draw the demand and supply curves, we need w on the left side:

Rearranging the demand function:

$$w = \frac{N^D - a}{b}$$

Rearranging the supply function:

$$w = \frac{N^S - c}{f}$$

Demand and Supply Functions

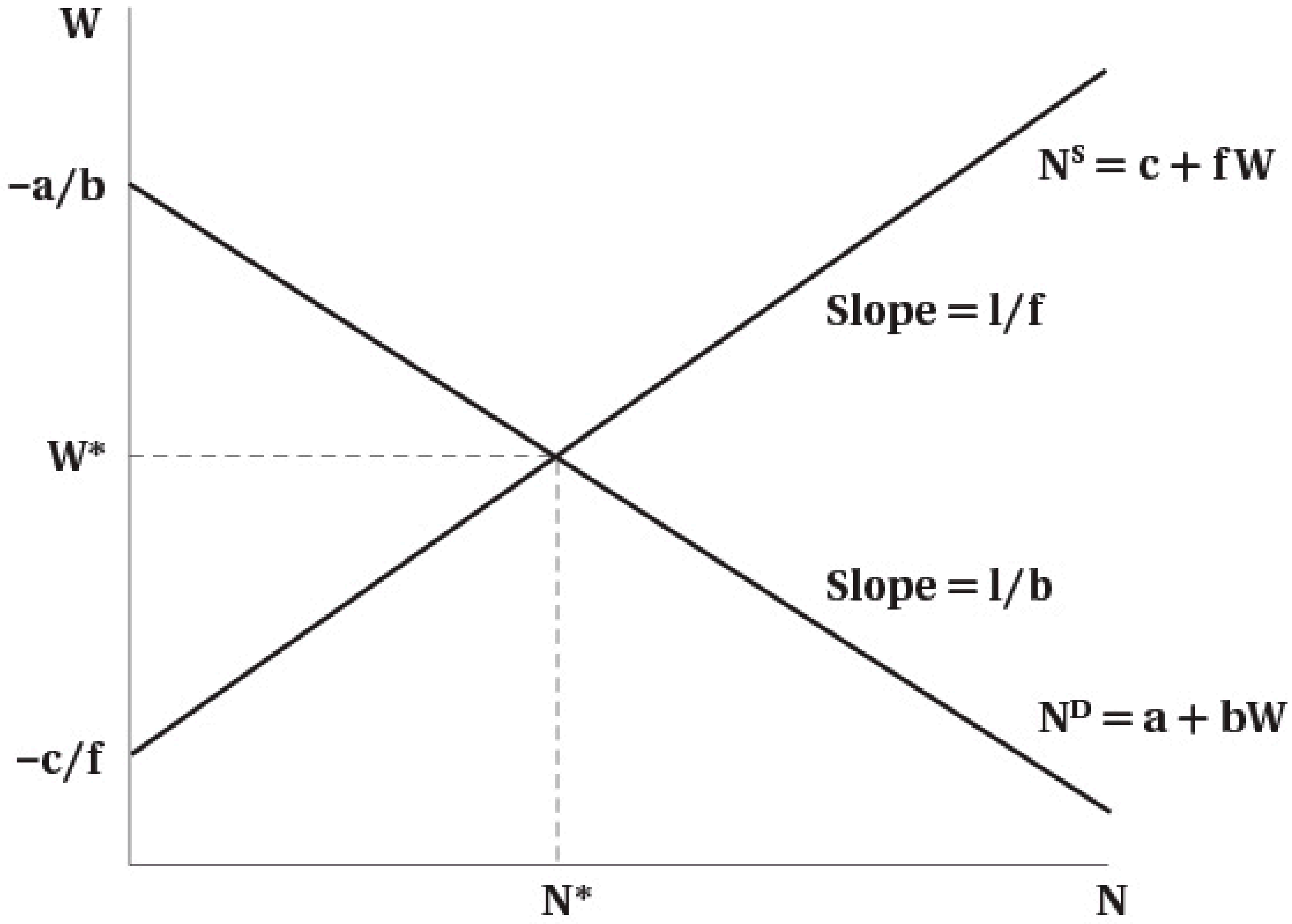


FIGURE 7.4 Linear Supply and Demand Functions

Payroll Tax

- We can use this model to analyze various policies
- One common policy is a payroll tax on employers
  - Examples: Canada Pension Plan (CPP), Employment Insurance (EI)
- These are typically a percentage of wages paid by employers
- We will analyze a fixed amount per hour for simplicity: suppose the tax is T per hour
- This drives a “wedge” between the wage paid by employers and the wage received by workers:

$$w_r = w_e - T$$

Payroll Tax

The demand function is:

$$w_e = \frac{N^D - a}{b}$$

The supply function is:

$$w_r = \frac{N^S - c}{f}$$

Payroll Tax

Substitute into the equilibrium price equation and set quantity demanded equal to quantity supplied:

$$\frac{N^* - c}{f} = \frac{N^* - a}{b} - T$$

Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af - bfT}{b - f}$$

Payroll Tax

Equilibrium Wages Under Payroll Tax

Wage received by workers:

$$w_r^* = \frac{c-a}{b-f} - \frac{b}{b-f}T$$

Wage paid by employers:

$$w_e^* = \frac{c-a}{b-f} - \frac{f}{b-f}T$$

Payroll Tax

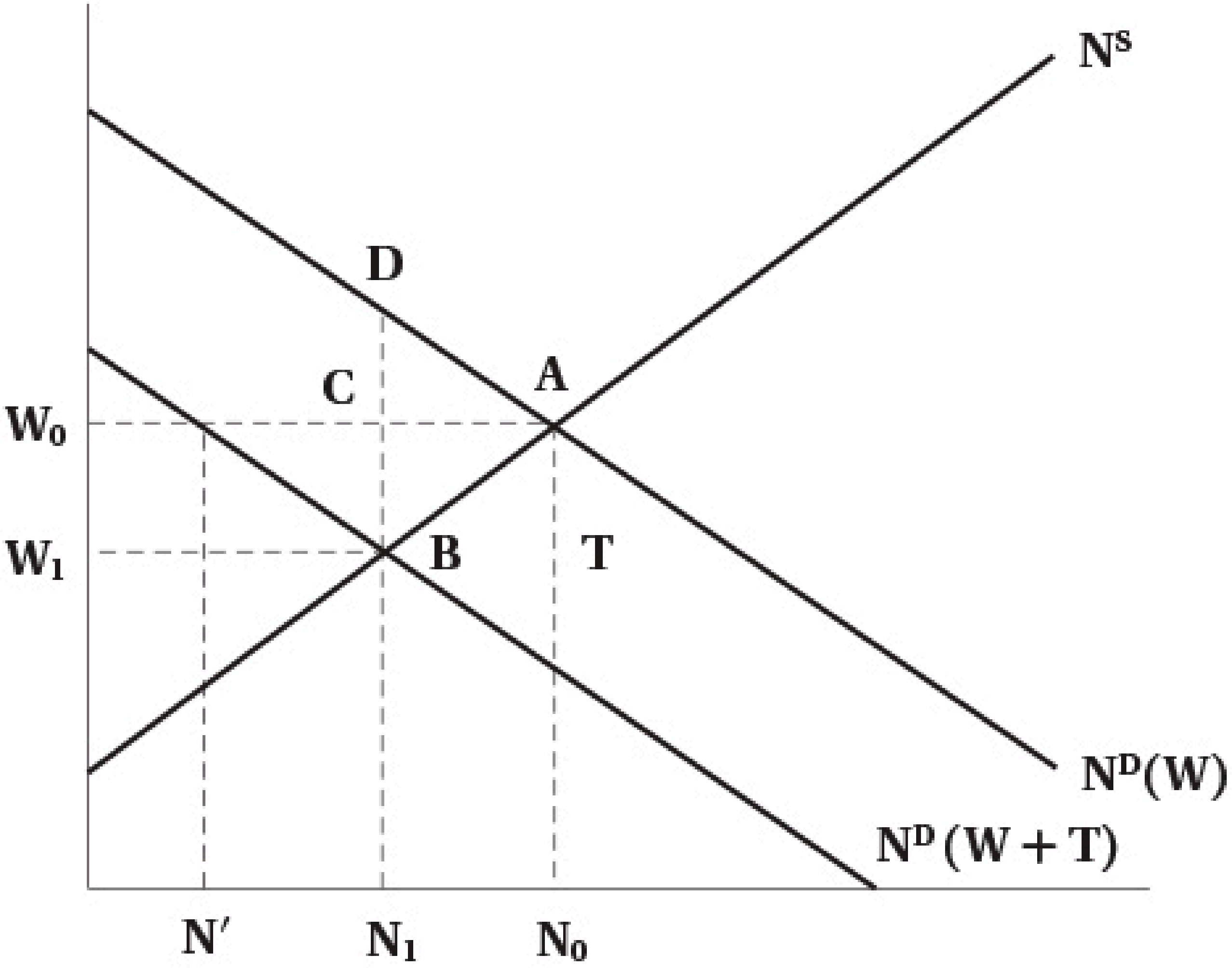


FIGURE 7.5 The Effect of a Payroll Tax on Employment and Wages

Payroll Tax

Tax Incidence

- Statutory incidence: who is legally responsible for paying the tax (here: firms)
- Economic incidence: who actually bears the burden of the tax (here: shared between firms and workers)

Economic incidence depends on elasticities of supply and demand:

- More elastic supply → workers bear less of the tax burden
- More elastic demand → firms bear less of the tax burden

Evidence

Empirical Findings on Payroll Taxes

- In the long run, workers bear most of the burden through lower wages
- Employment effects are small

Monopsony in the Labour Market

Introduction

> **Monopsony**
>
> Monopsony: a single buyer with market power in the labour market
>
> - Contrast with monopoly: single seller with market power in the goods market

Some examples of monopsony in the labour market:

- Company/factory towns
- Hospital labour markets
- University professor labour markets
- Public sector in small areas

In all cases, there is a large buyer of labour — giving the buyer power to set wages, affecting both wages and employment

Simple Monopsony

- In a monopsony, the firm faces the market supply curve for labour
  - Supply is upward sloping
  - Contrast to perfect competition where supply is horizontal
  - To hire more workers, the firm must pay a higher wage

Marginal Cost of Labour

The firm's marginal cost of labour is above the supply curve

- To hire an additional worker, the firm must raise the wage for **all** workers
- MC = cost of additional worker + increase in wage paid to existing workers
- Analogous to how monopolists face MR below demand

Simple Monopsony

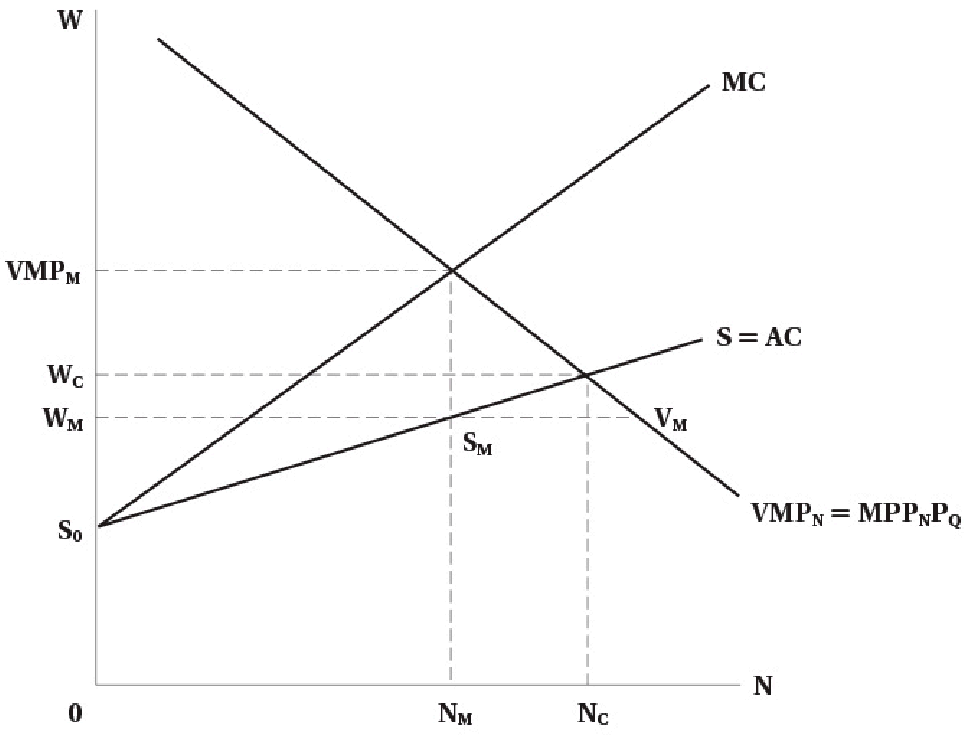


FIGURE 7.6 Monopsony

- Firm demand is its value of marginal product
- Faces upward sloping supply curve
- MC is above supply
- Optimal labour is where MC = VMP
- Wage comes from supply curve at that quantity of labour

Simple Monopsony

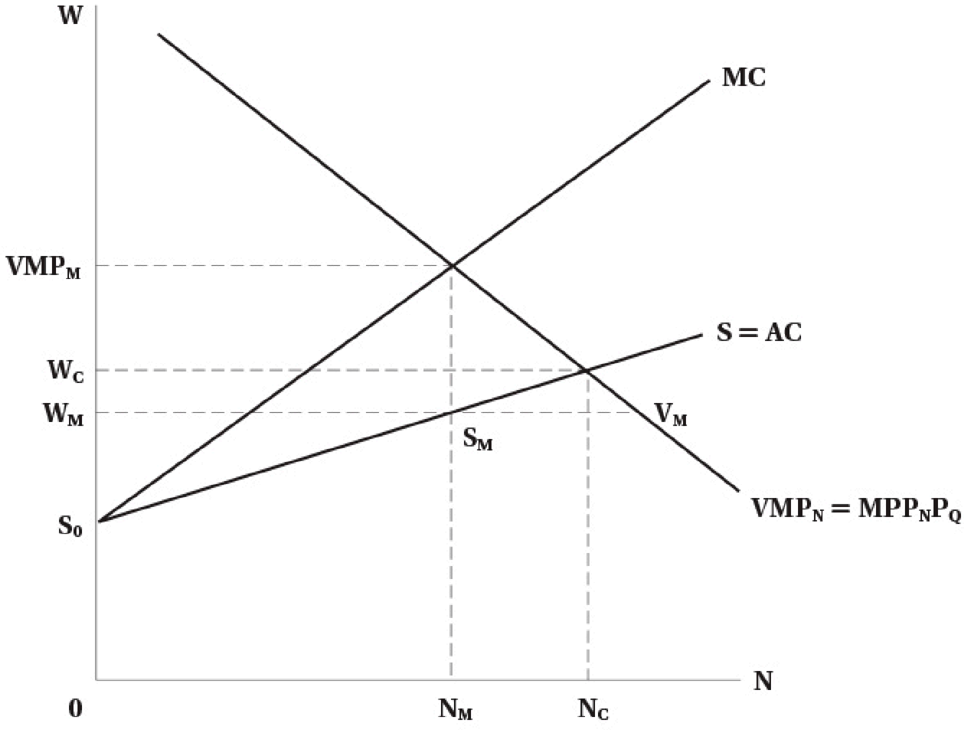


FIGURE 7.6 Monopsony

Compared to competitive equilibrium, monopsony results in:

- Wage below VMP
- Wage below competitive wage
- Employment below competitive level

Monopsony profit = $(VMP_m - W_m) \times N_m$

- Wage bill is $W_m \times N_m$
- Revenue contributed by workers is $VMP_m \times N_m$

Simple Monopsony

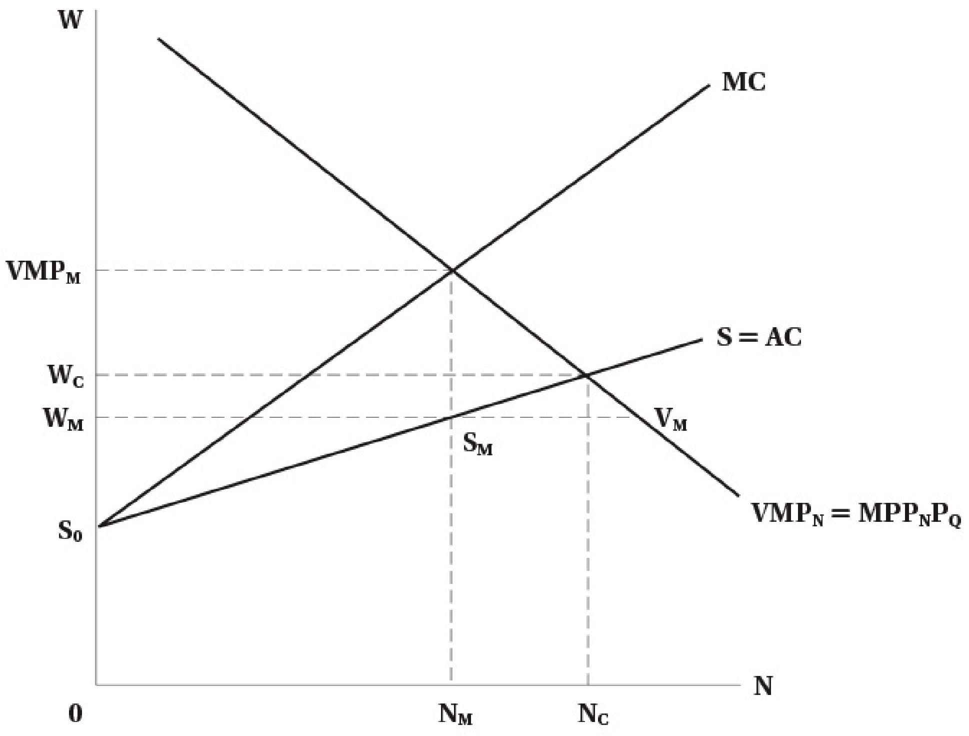


FIGURE 7.6 Monopsony

Monopsonies also leave jobs vacant:

- They restrict employment to maintain low wages
- This creates inefficiency in the labour market
- Vacancies are $V_m - S_m$

At wage W_m:

- VMP is above wage
- Firms would like to hire more
- But to hire, must raise wages — so it is optimal to not hire more

Characteristics of Monopsonies

Monopsonies tend to be:

- More prevalent in the short run
  - In the long run, workers can move to find other jobs
  - In the short run, frictions may prevent this
- Large relative to the labour market
  - Does not have to be a big firm, just large relative to the market
  - Example: a restaurant in a seasonal tourist town
- Associated with workers who have specialized skills
  - May naturally be few employers for specialized labour
  - Like academia, sports teams

Characteristics of Monopsonies

Note

Wages under monopsony are lower than competitive equilibrium, but not necessarily low in an absolute sense

- Example: sports players are not paid low wages, but market power makes them lower than they would be otherwise

Wage Differentiation

- In previous analysis, we assumed all workers are the same (single supply curve)
- This leaves some surplus for workers
  - There are workers willing to work for less than the equilibrium wage
- A firm can capture that surplus by differentiating wages
  - Paying different workers different wages based on their reservation wage

Wage Differentiation

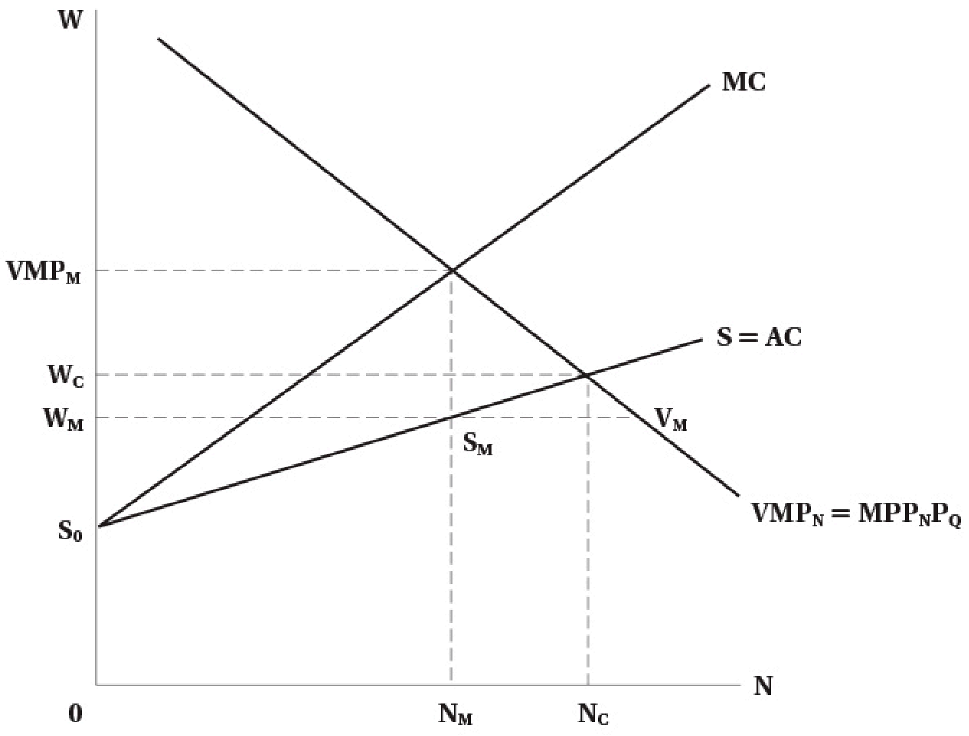


FIGURE 7.6 Monopsony

If the firm can pay different wages:

- It can pay each worker their reservation wage
- MC curve becomes equal to supply curve
- Firm will hire up to competitive level
  - It no longer has to increase wage for all workers to hire more

To do this, firms try to make wages secret:

- Prevent workers from knowing what others are paid
- Could use non-wage benefits for specific workers

Evidence of Monopsonies

Pro sports:

- Baseball: free agency increased wages significantly — increases competition among teams
- Hockey: not much evidence of monopsony power

Academia and nursing:

- Some evidence of monopsony power
- In Canada, a bargaining model may be more appropriate given these labour markets are unionized

> **Note**
>
> Some academics argue that monopsonies are relatively common — firms can set wages only if they have some market power, and evidence suggests this is widespread

Minimum Wages

Introduction

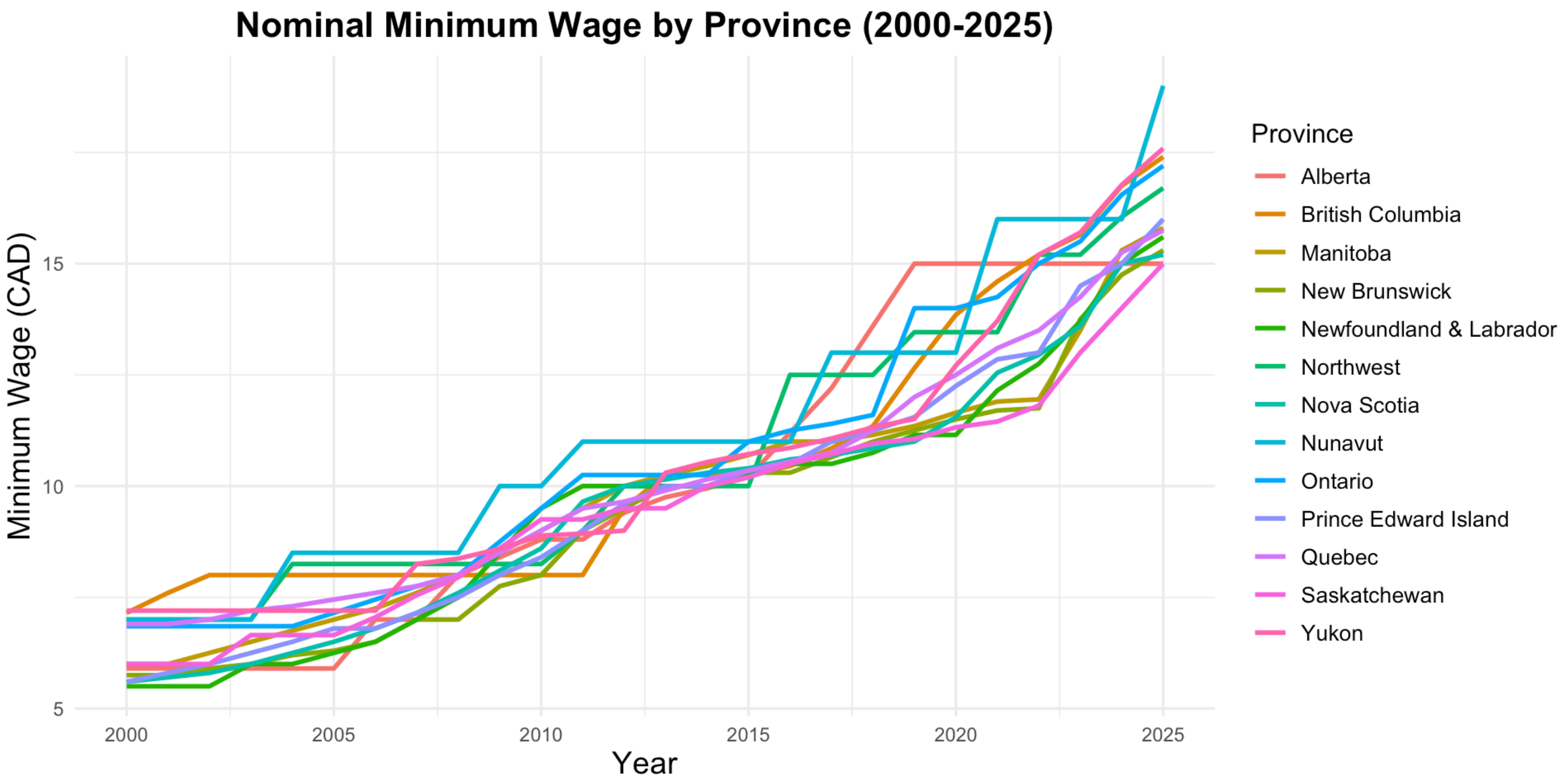
Minimum Wage

A minimum wage sets a legal floor on wages that employers must pay workers

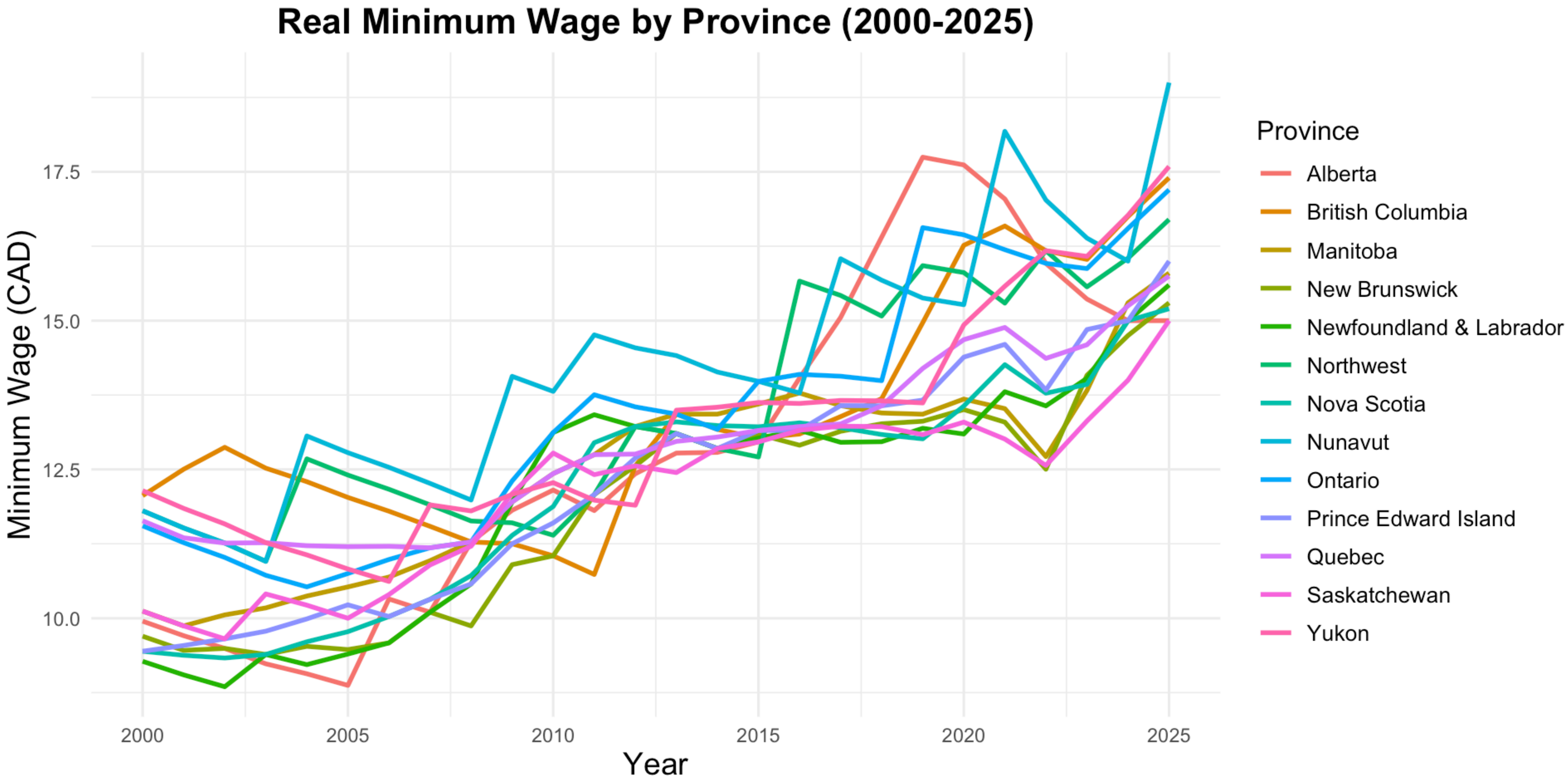
- Intended to raise wages of low-wage workers, reduce poverty, increase standard of living

- Standard economic theory predicts that minimum wages lead to unemployment
  - Because they set a wage above the equilibrium wage
  - Quantity of labour supplied exceeds quantity demanded
- But statistical evidence is much more mixed

Minimum Wages by Province



Minimum Wages by Province



Minimum Wages in Canada and USA

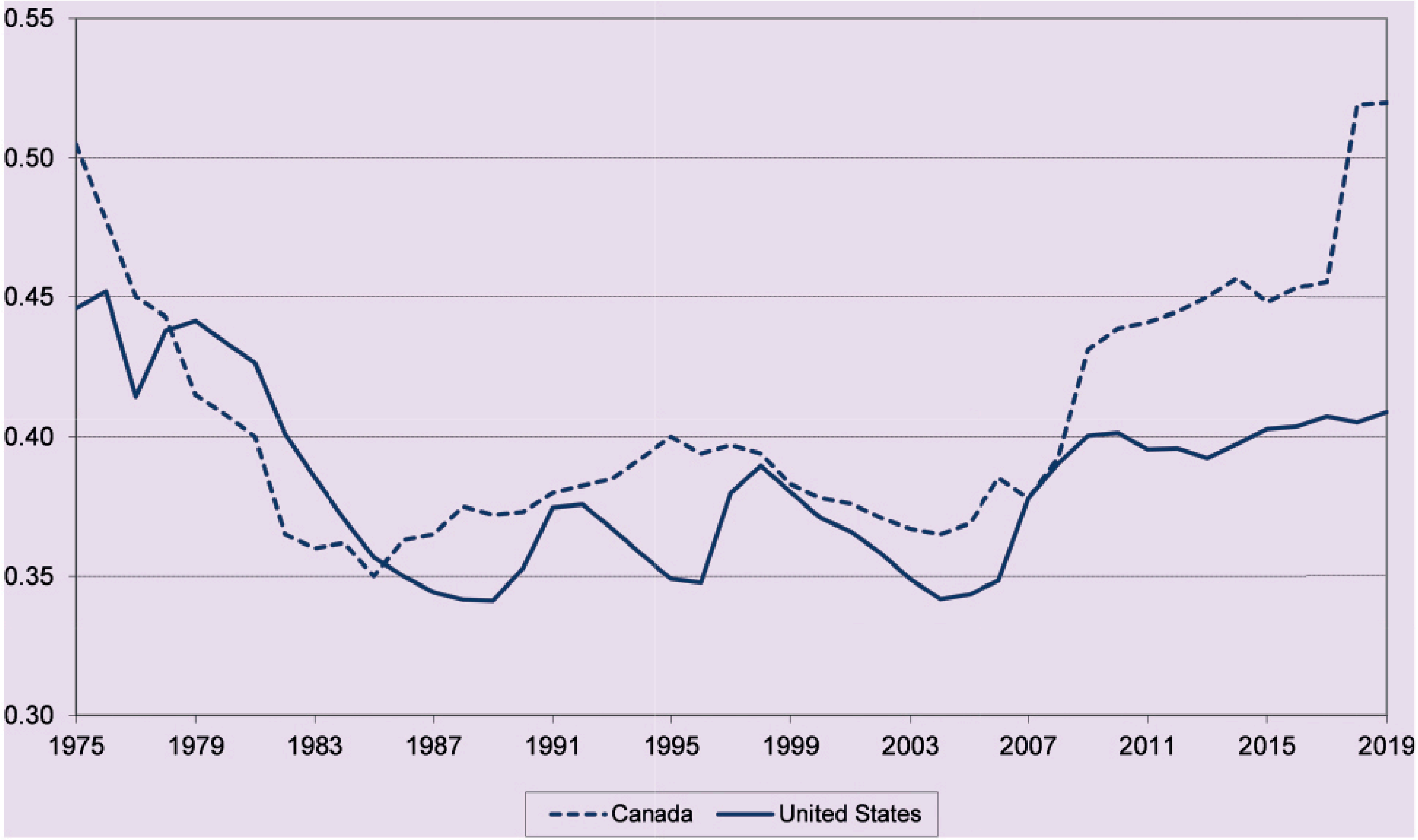


FIGURE 7.7 The Ratio of Minimum Wages to Average Wages, Canada and the United States, 1975–2019

Minimum Wages in Competitive Labour Market

In a competitive labour market:

- Market demand is downward sloping
- Market supply is upward sloping

A minimum wage acts as a wage floor set above the competitive equilibrium wage:

> **Warning**
>
> A binding minimum wage in a competitive market creates excess supply of labour, leading to unemployment

Minimum Wages in Competitive Labour Market

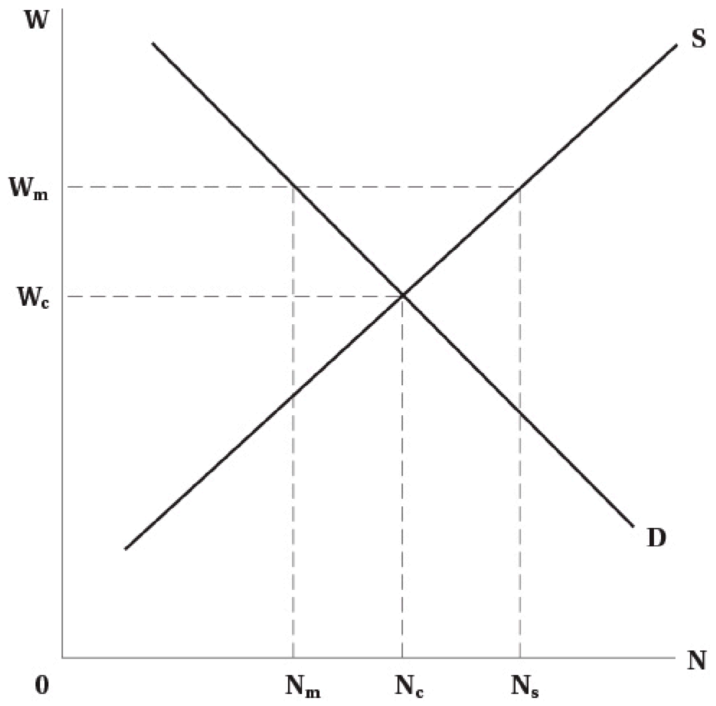


FIGURE 7.8 Minimum Wage

- Minimum wage W_m is above competitive level W_c
- Creates excess supply of labour
- Unemployment = $N_s - N_m$
  - Previously employed, now not: $N_c - N_m$
  - New entrants drawn in by higher wage: $N_s - N_c$

Minimum Wages in Competitive Labour Market

- Employment effects depend on elasticity of demand
  - More elastic demand → larger employment effects
- In the real world, industries with minimum wage workers tend to have elastic demands
  - Lots of substitutes for low-wage labour
  - Labour costs are a large share of total costs
- Nevertheless, we will see that minimum wages may not affect employment much in practice

Minimum Wages in Monopsony

Key Result

In a monopsonistic labour market, minimum wages can increase both wages and employment, depending on where the minimum wage is set

Minimum Wages in Monopsony

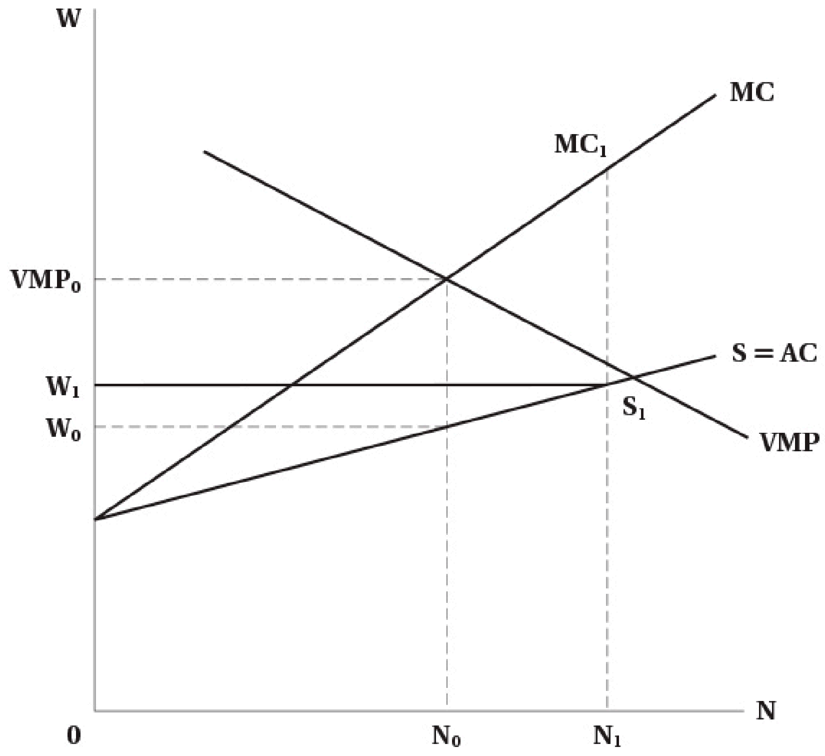


FIGURE 7.9 Monopsony and a Minimum Wage

- Prior to minimum wage: wage is W_0, employment is N_0
- Minimum wage set above W_0 but below competitive wage
- MC curve = minimum wage up to S_1, then jumps to previous MC curve
- Optimize where MC = VMP
- Employment increases to N_1

Minimum Wages in Monopsony

Why would minimum wages cause employment to increase?

- Previously, firm had to increase wages to attract more labour
- With minimum wage, it does not
  - Everyone is paid the same wage
  - Wage increases no longer inhibit employment growth
- Effectively, minimum wage lowers marginal cost

This may be partly why we do not see large employment effects of minimum wages

> **Note**
>
> Note that monopsonists do not like minimum wages — they reduce market power, increase costs, and reduce profits

Offsetting Factors for Minimum Wage

As usual, in models we hold all else equal — but in reality, many things change at once:

- Demand changes may accompany minimum wage increases
  - E.g., fiscal stimulus increasing demand for goods and labour
- Technology changes may follow
  - May change demand for labour
- Firms might choose to absorb costs
  - Reducing profits rather than cutting employment

Evidence

- Minimum wages are one of the most researched areas in labour economics
- Large amounts of evidence going back at least 50 years
- Early studies looked at time trends in employment and minimum wages
  - Subject to a lot of bias — things often trend together over time without direct causation
- Research in the early 1990s used better methods
  - Natural experiments
  - Difference-in-differences

Evidence

Card and Krueger (1994)

- Looked at fast food restaurants in New Jersey (raised minimum wage) and Pennsylvania (did not)
- Used difference-in-differences to estimate employment effects
- Found no negative employment effects from the minimum wage increase
- Has been reevaluated several times — conclusion remains that employment effects are small or non-existent

Evidence

In Canada:

- Minimum wages increase youth unemployment
  - Studies looked at minimum wage changes across provinces in the 1990s
- Some Canadian studies look deeper at labour market dynamics
  - In any month, roughly 6% of employed people separate, 8% of non-employed people are hired
  - Evidence shows minimum wages reduce both hiring and separations
  - Net effect on employment is approximately zero

Evidence

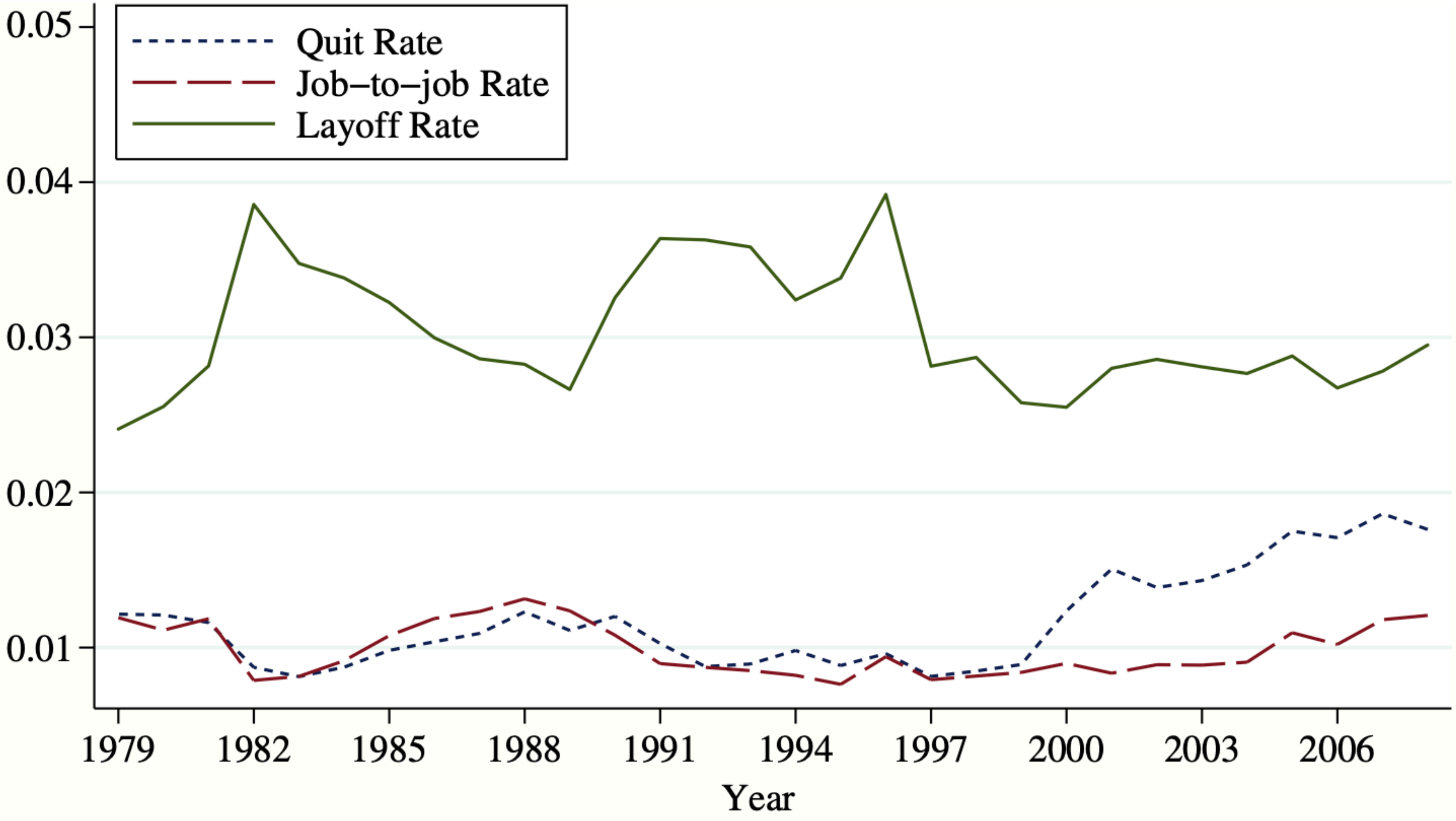


Fig. 4. *Layoff and Quit Rates, Low Skilled,* 1979–2008

Evidence

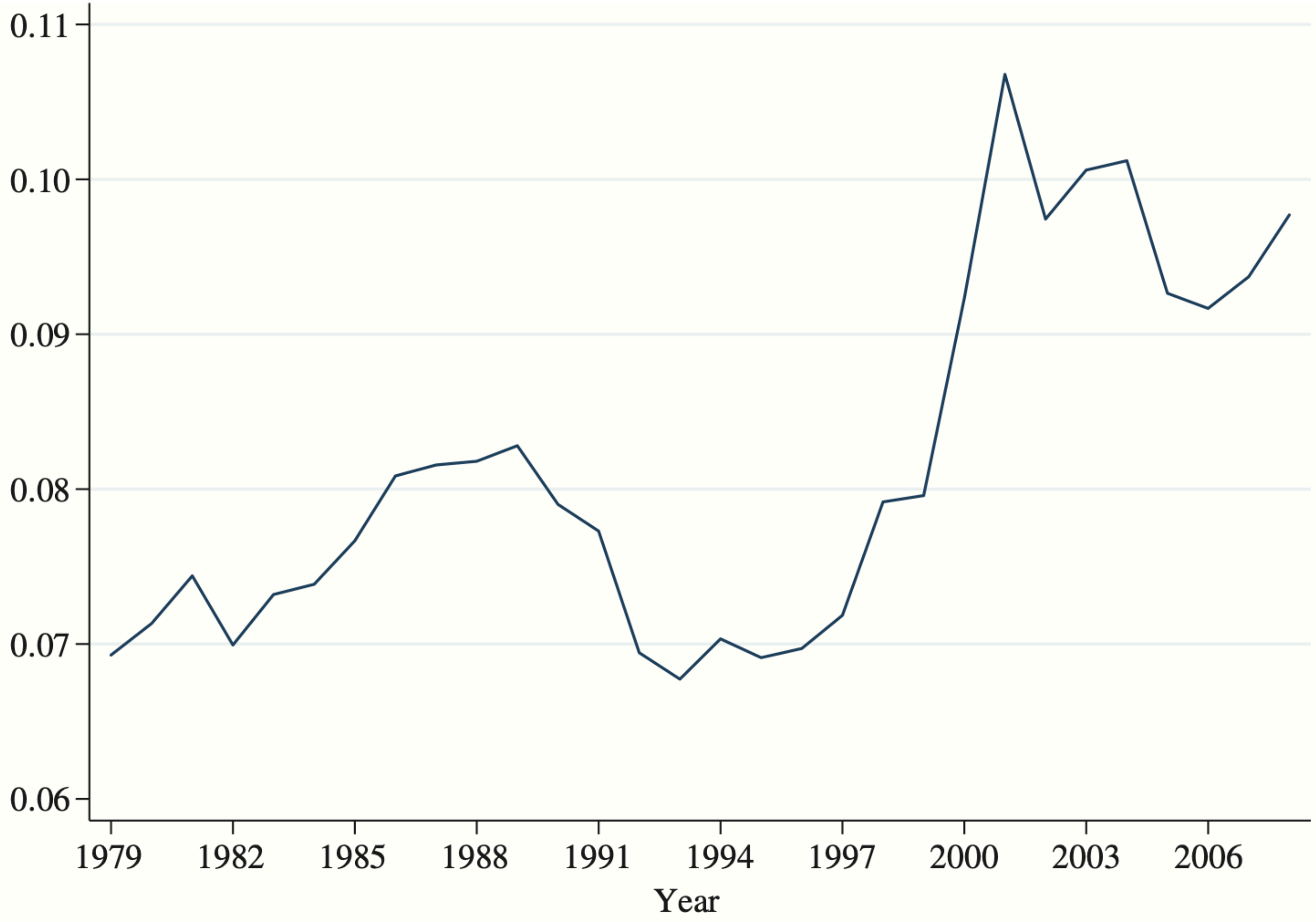


Fig. 5. *Hiring Rates, Low Skilled,* 1979–2008

Evidence

Warning

Minimum wages are generally not effective at reducing poverty

- Many poor individuals do not work
- Many people on minimum wages are not poor (e.g., youth living in high-income families)

Both Canadian and US studies show no link between minimum wages and poverty reduction

Summary

Summary

- We analyzed how wages and employment are determined in a single labour market
- Market power in the goods market reduces both wages and employment
- Market power in the labour market (monopsony) reduces wages and employment
- Minimum wages may reduce employment in competitive markets
- Minimum wages may increase employment in monopsonistic markets
- Empirical evidence shows minimum wages have small effects on employment